

Qn	Suggested Solutions
1(a)	$R = \frac{\rho l}{A}$ $\rho = \frac{RA}{l}$ $= \left(\frac{P}{I^2} \right) \frac{A}{l}$ $= \left(\frac{E}{I^2 t} \right) \frac{A}{l}$ $\text{unit of } \rho = \frac{\text{unit of } mghA}{\text{unit of } I^2 t l}$ $= \frac{(\text{kg})(\text{m s}^{-2})(\text{m})(\text{m}^2)}{\text{A}^2 (\text{s})(\text{m})}$
	unit of $\rho = \text{kg m}^3 \text{ s}^{-3} \text{ A}^{-2}$
(b)	Assume power output of phone charger $\sim 10 \text{ W}$ Typical USB charging voltage $\sim 5 \text{ V}$ $I = \frac{P}{V}$ $= \frac{10}{5}$ $= 2 \text{ A}$
(c)	$R = \frac{V}{I}$ $= \frac{5.02}{0.038}$ $= 132.105 \Omega$
	$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$ $\frac{\Delta R}{132.105} = \frac{0.01}{5.02} + \frac{0.001}{0.038}$ $\Delta R = 3.7$ $\approx 4 \Omega$
	$R \pm \Delta R = (132 \pm 4) \Omega$

Anglo-Chinese Junior College

H2 (9749) Physics

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